

JUAN VASCONEZ

PHYSICS, DATA SCIENCE, AI

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PROFESSIONAL SUMMARY

Physics undergraduate with strong expertise in data science, statistical analysis, and programming. Passionate about leveraging AI and data analysis for social good and environmental solutions. Experienced in developing scalable solutions using Python, SQL, and R while collaborating with AI experts to deliver high-quality products.

EDUCATION: PHYSICS MAJOR

Yachay Tech University | Expected Graduation: DECEMBER 2026 | ECUADOR

A degree in physics that provides me with a solid understanding of the fundamental principles governing the universe. Skills in analytical problem solving and experience in both theoretical and experimental approaches. The degree reflects my commitment to precision, logical thinking, and a great interest in unraveling the complexities of the physical world.

WORK EXPERIENCE

Data Scientist | Diversa | January 2025 - Present

Junior Data Scientist | Diversa | January 2024 - January 2025

- Collaborate with experts in AI and Data Analysis to develop human-centered and environmentally conscious solutions
- Implement scalable data solutions using Python, SQL, and multiple libraries for client projects.
- AI Agents implementation and full stack applications.
- Design and develop ethical AI applications focused on responsible data practices.
- Analyze complex datasets to extract actionable insights for business decision-making.
- UI/UX Design.
- Generative UI implementations.

Security Systems Manager

Orion Seguridad | 2019, 2024

- Managed and optimized a comprehensive security camera network of 2000+ video recorders on Linux servers
- Developed streamlined workflows that improved security response time by 30%
- Conducted live monitoring of security footage across a national university campus
- Maintained and troubleshoot Linux-based server infrastructure for continuous operation

High School Physics & Mathematics Tutor

Unidad Educativa Fiscomisional Salesiana Sanchez y Cifuentes | 2023

- Developed and delivered targeted lessons to improve students logical analysis and critical thinking skills
- Created customized learning materials for mathematics, chemistry, and physics
- Mentored students through complex problem-solving techniques with measurable improvement in test scores

HABILITIES

PROGRAMMING SKILLS:

- Python ★ ★ ★ ★ ★
- AI & ML on Python ★ ★ ★ ★
- SQL, PostgreSQL ★ ★ ★
- AI Agents ★ ★
- Scientific Programming ★ ★ ★ ★ ★
- Data Analysis & Visualization ★ ★ ★ ★ ★
- Frontend Development ★ ★
- Backend Development ★ ★ ★ ★

PROFESSIONAL SKILLS:

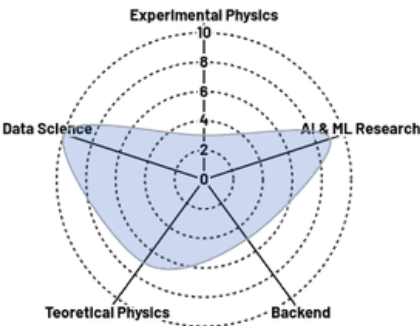
- Scientific Writing & Research ★ ★ ★ ★
- Process Optimization ★ ★ ★ ★ ★
- Prompt Engineering ★ ★ ★ ★ ★
- Content Creation ★ ★ ★
- Video and Audio Editing ★ ★ ★ ★

TOOLS & TECHNOLOGIES:

- Linux (Administration & Kernel) ★ ★ ★ ★
- Version Control (Git) ★ ★ ★
- Quantum Computing (Qiskit) ★ ★ ★ ★
- Deployment techniques. ★ ★ ★
- HPC Computing ★ ★ ★

LANGUAGES

- English (Advanced) ★ ★ ★ ★ ★
- Spanish (Native) ★ ★ ★ ★ ★



PERSONAL WORK

- Clubs
 - Computational Physics | President | Yachay Tech | 1 Year
 - Basketball Club | President | Yachay Tech | 2 Years
 - CAS IEEE Chapter | President | Yachay Tech | 1 Year
 - IEEE Student Chapter | Web Master | Yachay Tech | 1 Year
- Comitees
 - EPIC V Organizing Committee | 2025 | Investigation Group Latitud 0
 - Physics and Nano-technology Asociation of Students | President | 2025
 - Olympic Tournament Basketball | 2024 & 2025 | Organizer | Yachay Tech
- Schools
 - EPIC IV Scientific Programming | 2024 | UIDE
 - Internation School for Young Astronomers (ISYA) | 2025 | Yachay Tech
 - Quantum Computing School | 2023 | Yachay Tech
- Personal Projects
 - Astrophysical Images Segmentation with Diff Cut | Artificial Intelligence | 2025
 - Computer Vision Approaches for Automated Skin Cancer Detection | 2025
 - Hand Movement prediction by ML with EMG signals | 2025
 - Waves and Oscillations | Teaching Assistant | 2025

PUBLICATIONS

- Integrating Artificial Intelligence (AI) and Feminist Urban Design: A Comparative Study of Public Spaces in South America and Southeast Asia. | Waiting for Publication | Diversa
- Spectral Methods for the Navier Stokes Equations | Poster | NanoPhys Day V4

REFERENCES

- Diana Mosquera | Co-founder & AI Research | Diversa | diana@diversa.studio
- Edward Gaibor | Security Head | Uniandes | edwardinversiones@hotmail.com